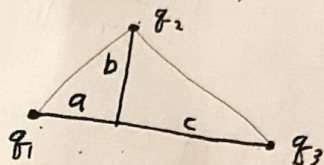


one form of Schrodinger equation.

5. Calculate the energy of assembly of the system of charges. q_1 is brought from infinity first, then q_2 , then q_3 , then q_4 . Give a detailed symbolic solution. Write appropriate distances in terms of labeled distances.



$$U_2 = U_{21} = \frac{kq_2q_1}{r_{21}} = \frac{kq_2q_1}{\sqrt{a^2 + b^2}}$$

$$U_3 = U_{31} + U_{32} = \frac{kq_3q_1}{r_{31}} + \frac{kq_3q_2}{r_{32}} = \frac{kq_3q_1}{a+c} + \frac{kq_3q_2}{\sqrt{b^2 + c^2}}$$

$$U_{\text{assembly}} = U_2 + U_3$$

Ask about

- a) Electric field
- b) Electric potential
- c) Electric Force on q_2
- d) Change in Electric PE.

(H 23 sec 5)

$$W = \int F \cos \theta dx = -\Delta U$$

$$W = \int \vec{F} \cdot d\vec{s} = -\Delta U$$

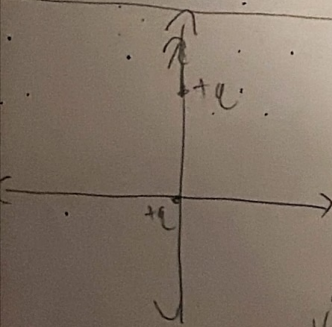
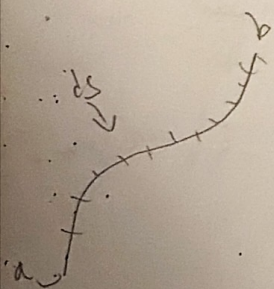
$$F dx = -dU \quad F = -\frac{dU}{dx} \quad \text{(H 7 sec 4)}$$

Now let $v = U(x, y, z)$

Then find ∇U

$$\frac{-W}{q} = \int \frac{\vec{F}}{q} \cdot d\vec{s}$$

$$E \cos \theta = \cos \theta =$$



[2] $V_f - V_i = -\int_{r_i}^{r_f} \left(\frac{kq}{r^2} \right) dr$ [1] $\Delta V = -\int_{r_i}^{r_f} E \cdot dr$

[3] They point in same direction, so V_f when they're parallel and thus $\theta = 0^\circ$, max $F_P = \infty$ is at product.

[4] so $V_f - V_i = -kq \int_{r_i}^{r_f} r^{-2} dr = -kq \left(\frac{1}{r_f} - \frac{1}{r_i} \right)$
 so $V_f = 0$ $r_f = \infty$, so $\frac{1}{\infty} = 0$, so $kq \left(-\frac{1}{r_i} \right)$ negatives cancelled.
 $V_i = -\frac{kq}{r}$ $r_f = \infty$, so $\frac{1}{\infty} = 0$, so $kq \left(-\frac{1}{r_i} \right)$ negatives cancelled.